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Leadless cardiac pacemaker with secondary fixation capability

Abstract

The invention relates to leadless cardiac pacemakers (LBS), and elements and methods by which they affix to the heart. The invention relates particularly to a secondary fixation of leadless pacemakers which also include a primary fixation. Secondary fixation elements for LBS's may passively engage structures within the heart. Some passive secondary fixation elements entangle or engage within intraventricular structure such as trabeculae carneae. Other passive secondary fixation elements may engage or snag heart structures at sites upstream from the chamber where the LBS is primarily affixed. Still other embodiments of passive secondary fixation elements may include expandable structures.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) .[.This application is a continuation of U.S. patent application Ser. No. 12/698,969, now U.S. Pat. No. 8,527,068, filed Feb. 2, 2010,

which application claims the benefit under 35 U.S.C. 119 of U.S. Provisional Patent Application No. 61/149,176, filed Feb. 2, 2009, titled "Leadless Cardiac Pacemaker With Secondary Fixation Capability", and U.S. Provisional Patent Application No. 61/155,038, filed Feb. 24, 2009, titled "Leadless Cardiac Pacemaker With Secondary Fixation Capability". These applications are herein incorporated by reference in their entirety.]. .Iadd.This application is a reissue application of U.S. Pat. No. 9,272,155 granted on Mar. 1, 2016, which issued from U.S. application Ser. No. 13/967,180 filed on Aug. 14, 2013, which is a continuation of U.S. patent application Ser. No. 12/698,969, filed Feb. 2, 2010, now U.S. Pat. No. 8,527,068, which application claims the benefit under 35 U.S.C. 119 of U.S. Provisional Patent Application No. 61/149,176, filed Feb. 2, 2009, titled "Leadless Cardiac Pacemaker With Secondary Fixation Capability", and U.S. Provisional Patent Application No. 61/155,038, filed Feb. 24, 2009, titled "Leadless Cardiac Pacemaker With Secondary Fixation Capability", and these applications are herein incorporated by reference in their entirety. .Iaddend.

INCORPORATION BY REFERENCE

(1) All publications, including patents and patent applications, mentioned in this specification are herein incorporated by reference in their entirety to the same extent as if each individual publication was specifically and individually indicated to be incorporated by reference.

FIELD OF THE INVENTION

(2) The present invention relates to leadless cardiac pacemakers, and more particularly, to features and methods by which they are affixed within the heart.

BACKGROUND OF THE INVENTION

(3) Cardiac pacing by an artificial pacemaker provides an electrical stimulation of the heart when its own natural pacemaker and/or conduction system fails to provide synchronized atrial and ventricular contractions at rates and intervals sufficient for a patient's health. Such antibradycardial pacing provides relief from symptoms and even life support for hundreds of thousands of patients. Cardiac pacing may also provide electrical overdrive stimulation to suppress or convert tachyarrhythmias, again supplying relief from symptoms and preventing or terminating arrhythmias that could lead to sudden cardiac death.

(4) Cardiac pacing by currently available or conventional pacemakers is usually performed by a pulse generator implanted subcutaneously or sub-muscularly in or near a patient's pectoral region. Pulse generator parameters are usually interrogated and modified by a programming device outside the body, via a loosely-coupled transformer with one inductance within the body and another outside, or via electromagnetic radiation with one antenna within the body and another outside. The generator usually connects to the proximal end of one or more implanted leads, the distal end of which contains one or more electrodes for positioning adjacent to the inside or outside wall of a cardiac chamber. The leads have an insulated electrical conductor or conductors for connecting the pulse generator to electrodes in the heart. Such electrode leads typically have lengths of 50 to 70 centimeters.

(5) Although more than one hundred thousand conventional cardiac pacing systems are implanted annually, various well-known difficulties exist, of which a few will be cited. For example, a pulse generator, when located subcutaneously, presents a bulge in the skin that patients can find unsightly, unpleasant, or irritating, and which patients can subconsciously or obsessively manipulate or "twiddle". Even without persistent manipulation, subcutaneous pulse generators can exhibit erosion, extrusion, infection, and disconnection, insulation damage, or conductor breakage at the wire leads. Although sub-muscular or abdominal placement can address some concerns, such placement involves a more difficult surgical procedure for implantation and adjustment, which can prolong patient recovery.

(6) A conventional pulse generator, whether pectoral or abdominal, has an interface for connection to and disconnection from the electrode leads that carry signals to and from the heart. Usually at

least one male connector molding has at least one terminal pin at the proximal end of the electrode lead. The male connector mates with a corresponding female connector molding and terminal block within the connector molding at the pulse generator. Usually a setscrew is threaded in at least one terminal block per electrode lead to secure the connection electrically and mechanically. One or more O-rings usually are also supplied to help maintain electrical isolation between the connector moldings. A setscrew cap or slotted cover is typically included to provide electrical insulation of the setscrew. This briefly described complex connection between connectors and leads provides multiple opportunities for malfunction.

(7) Other problematic aspects of conventional pacemakers are enumerated in the related applications, many of which relate to the separately implanted pulse generator and the pacing leads. By way of another example, the pacing leads, in particular, can become a site of infection and morbidity. Many of the issues associated with conventional pacemakers are resolved by the development of a self-contained and self-sustainable pacemaker, or so-called leadless pacemaker, as described in the related applications cited above.

(8) Self-contained or leadless pacemakers or other biostimulators are typically fixed to an intracardial implant site by an actively engaging mechanism such as a screw or helical member that screws into the myocardium. Examples of such leadless biostimulators are described in the following publications, the disclosures of which are incorporated by reference: (1) U.S. application Ser. No. 11/549,599, filed on Oct. 13, 2006, now U.S. Pat. No. 8,457,742, entitled "Leadless Cardiac Pacemaker System for Usage in Combination with an Implantable Cardioverter-Defibrillator"; (2) U.S. application Ser. No. 11/549,581 filed on Oct. 13, 2006, entitled "Leadless Cardiac Pacemaker", and published as US2007/0088396A1 on Apr. 19, 2007; (3) U.S. application Ser. No. 11/549,591, filed on Oct. 13, 2006, entitled "Leadless Cardiac Pacemaker System with Conductive Communication" and published as US2007/0088397A1 on Apr. 19, 2007; (4) U.S. application Ser. No. 11/549,596 filed on Oct. 13, 2006, now U.S. Pat. No. 8,352,025, entitled "Leadless Cardiac Pacemaker Triggered by Conductive Communication"; (5) U.S. application Ser. No. 11/549,603 filed on Oct. 13, 2006, now U.S. Pat. No. 7,937,148, entitled "Rate Responsive Leadless Cardiac Pacemaker"; (6) U.S. application Ser. No. 11/549,605 filed on Oct. 13, 2006, now U.S. Pat. No. 7,945,333, entitled "Programmer for Biostimulator System"; (7) U.S. application Ser. No. 11/549,574, filed on Oct. 13, 2006, now U.S. Pat. No. 8,010,209, entitled "Delivery System for Implantable Biostimulator"; and (8) International Application No. PCT/US2006/040564, filed on Oct. 13, 2006, entitled "Leadless Cardiac Pacemaker and System" and published as WO07047681A2 on Apr. 26, 2007.

(9) The site of attachment of leadless biostimulators is physically reinforced by a foreign body response that results in the growth of fibrotic tissue that further secures the leadless biostimulator at the attachment site. A high degree of success of attachment by such an approach notwithstanding, the potential of detachment of the leadless biostimulator from the implant site would represent an immediately serious event, as for example, a pacemaker lost from the right ventricle can exit the heart via the pulmonic valve and lodge in the lung. Leadless or self-contained biostimulators would benefit from mechanisms and methods for "secondary fixation" of the device within the heart, or more generally, features that in the event of failure of the primary fixation to the implant site would prevent escape of the pacemaker into the circulation downstream from the heart.

SUMMARY OF THE INVENTION

(10) The invention relates to a leadless cardiac pacemaker, a device more generally referred to as a leadless biostimulator (LBS), which includes a primary fixation element and a secondary fixation element. The invention also relates to methods of implanting a biostimulator with such a secondary fixation feature, and more generally to methods for retaining a leadless biostimulator in the heart in the event that the biostimulator is dislodged from its site of primary fixation.

(11) With regard to embodiments of a leadless biostimulator with both primary and secondary fixation features, embodiments of the primary fixation element may be either active or passive;

active elements typically requiring an active engagement of the element to a portion of the heart on the part of the user implanting the LBS and/or an active or at least minimally invasive engagement of heart structure, and the passive embodiments not so-requiring.

(12) Embodiments of the secondary fixation element or assembly may also be characterized as active or passive. Exemplary embodiments of active forms of a secondary fixation assembly include an anchor and a tether, the tether connecting the LBS to the anchoring site, the anchoring being established by securely engaging or embedding within heart or vascular structure at the time of implantation. Various types of more passive forms of secondary fixation are also provided by the invention. Exemplary embodiments of passive forms of the secondary fixation element or assembly include features that engage or engage, snag, or entangle structural features within the heart and thereby prevent its escape downstream in the event that it becomes dislodged from the primary fixation site. Some embodiments of passive types of fixation include entangling elements connected to the LBS which become entangled in structural features within the heart chamber where the LBS is implanted. Other embodiments of passive types of fixation include heart structure-engaging elements that engage or snag heart structure that is located upstream from the site of primary fixation. Still other embodiments of passive types of fixation include expandable structures which, in their expanded configuration, have a minimal cross-sectional dimension that is greater than the diameter of the vascular exit pathway from the heart chamber in which the LBS is implanted.

(13) Embodiments of a leadless biostimulator typically include a primary fixation element adapted to affix the biostimulator to a primary fixation site on a heart wall within a heart chamber; and a downstream vascular escape prevention assembly adapted to prevent an escape of the biostimulator in the event of it being dislodged from the implant site in a chamber of the heart. Other components of the leadless biostimulator include a power source adapted to be disposed within a human heart chamber, an electrode in electrical communication with the power source and adapted to be placed in contact with tissue within the heart chamber, a controller adapted to be disposed within the heart chamber and to control delivery of electrical energy from the power source to the electrode. Some embodiments of the leadless biostimulator include a housing within which the power source, the electrode, and the controller are disposed. Some embodiments of the biostimulator may be adapted for implantation in the right ventricle or the left ventricle of the heart; in other embodiments, the biostimulator may be implanted in the left or right atrium of the heart.

(14) Some embodiments of a leadless biostimulator have a downstream vascular escape prevention assembly that includes one or more entangling elements adapted to entangle within heart structure at one or more secondary fixation sites within the chamber of the heart. In some of these embodiments, the one or more entangling elements may include any of tines, hooks, or chains. Typical embodiments of entangling elements are adapted to extend radially outward beyond the diameter of the biostimulator, particularly after the biostimulator is implanted. Some of the entangling element embodiments are at least 5 mm in length. Some of the entangling element embodiments extend outward from the biostimulator at a proximal-facing angle that ranges from about 10 degrees to about 90 degrees from the axis of the biostimulator. Some of the entangling element embodiments such as tines are configured as any of straight tines, curvilinear tines, or convoluted tines.

(15) Some of the entangling element embodiments are adapted to be rotatable with respect to the biostimulator, as for example, they may be mounted on a rotatable collar encircling the main axis of the biostimulator. Some of the entangling element embodiments are configured such that they are distally-collapsible around the periphery of the biostimulator. When collapsed, typical embodiments of collapsible entangling elements are configured to be substantially contained within a maximal diameter of the biostimulator, or add a minimal increment to such maximal diameter.

(16) As noted above, some embodiments of passive secondary fixation element engage heart structure upstream from the site of primary fixation. Thus, a leadless biostimulator that includes

this form of secondary fixation, in addition to the power source, electrode, controller, and primary fixation element, as summarized above, includes a downstream vascular escape prevention assembly adapted to prevent an escape of the biostimulator in the event of it being dislodged from the primary fixation site, such assembly including a heart structure engaging element adapted to passively engage or snag a heart structure as a secondary fixation site within the heart.

(17) Embodiments of this particular downstream vascular escape prevention assembly include a tail attached to the distal portion of the biostimulator. Such a tail may include a metallic core, and such core may further include a multifilar wrapping around the core. The tail may further include a polymeric coating, such as polyurethane or silicone. The tail may further include surface features or agents that discourage the attachment and colonization of the surface of the tail by cells, such as blood borne cells or cardiac tissue cells. Various embodiments of the tail may also include an atraumatic tip configured to prevent damage to cardiac tissue, and the tail may also include a feature that is adapted to enhance capturability of the LBS by a retrieval device.

(18) Still other embodiments of a leadless biostimulator that include a passive secondary fixation element may be summarized as a leadless biostimulator that includes a power source adapted to be disposed within a chamber of a human heart, an electrode in electrical communication with the power source and adapted to be placed in contact with tissue within the heart chamber, a controller adapted to be disposed within the heart chamber and to control delivery of electrical energy from the power source to the electrode, a primary fixation element adapted to affix the biostimulator to a primary fixation site on a heart wall within the heart chamber; and a downstream vascular escape prevention assembly adapted to prevent an escape of the biostimulator in the event of it being dislodged from the primary fixation site, the downstream vascular escape prevention assembly comprising an expandable structure. The expandable structure has a constrained or compressed configuration appropriate for deployment and an unconstrained default or radially expanded configuration. This radially-expanded configuration, in comparison to the compressed configuration, has a minimal cross-sectional linear dimension greater than a downstream vascular exit from the heart chamber. Some embodiments of the expandable structure formed at least in part from a shape memory material, such as Nitinol, whose properties confer on the structure an ability to return to a default or unconstrained configuration, after being forced or constrained into an alternative configuration.

(19) The leadless biostimulator, as just summarized, may further include a coupling mechanism that couples the expandable structure to a proximal portion of the biostimulator. In some particular embodiments, the coupling mechanism is adapted to provide freedom of movement between the expandable structure and the biostimulator, as for example the coupling mechanism may include a swivel, a tether, a universal joint, a combination of any of the preceding mechanisms, or any conventional mechanism that provides a freedom of movement between two coupled structures.

(20) Embodiments of the expandable structure of the leadless biostimulator, as summarized above, have a central longitudinal axis and may include struts disposed parallel to that longitudinal axis, which collectively form the expandable structure. These struts typically have a native bowed configuration, and they may further be adapted to be constrainable into a linear configuration without compromising their ability to return to a native configuration. In some embodiments, the bowed configuration is longitudinally symmetrical. In other embodiments, the bowed configuration is longitudinally asymmetrical, such that, for example, the asymmetrical bowed configuration is distributed such that a greater radial diameter is proximally-skewed.

(21) Some embodiments of a the leadless biostimulator have a downstream vascular escape prevention assembly that includes a tether and an anchor, the tether connecting the assembly and the anchor to each other, and the anchor adapted to anchor at a secondary attachment site. In these embodiments, the anchor may include any of a screw, a hook, a clip, a stent, a cage, or a barb to attach the biostimulator to the secondary attachment site. The attachment site to which the anchor plus tether embodiments of secondary fixation to which the anchor is adapted to affix may be any

of an intracardiac site, an intravascular site, or an extravascular site. In some embodiments, the intracardiac site is a septal wall of the heart. In other embodiments, the intravascular site is located within a vessel through which the biostimulator was delivered to the heart. Such vessels may include, for example, any of the femoral vein or the inferior vena cava. In some of these embodiments, the tether of the biostimulator is formed from two segments secured together with a clip. In other embodiments, an extravascular site may include the external periphery of a vessel through which the biostimulator was delivered to the heart. In these embodiments, the tether is typically adapted to be threaded through the vessel wall and to be attached to an anchor, the anchor including, by way of example, any of a partial cylinder, a plate, or a ball. In some anchor-plus-tether embodiments, the connection between the anchor and the tether, or between the tether and the biostimulator may include intervening or connective elements.

(22) In some embodiments of a leadless biostimulator, the anchor may include one or more electrodes for biostimulation, wherein the tether itself is electrically conductive. In some embodiments, the tether may include any of single strand wire, multistranded wire, monofilament suture thread, or multistrand suture thread. In some embodiments, a tether or any of the anchor itself, or entangling elements may include any of a biodegradable material or an antithrombogenic agent.

(23) Some embodiments of a leadless biostimulator may include one or more soluble coverings configured to encapsulate any of the primary fixation element or the secondary fixation element. Some embodiments of the soluble covering may include biocompatible materials, such as, merely by way of example, a polymer (such as polyvinylpyrrolidone), a protective sugar (such as mannitol), or a protective salt. In typical embodiments that make use of a soluble covering that is useful in deployment of the device, the soluble covering secures the secondary element in a collapsed configuration.

(24) As mentioned above, embodiments of the invention also include a method for retaining a leadless intracardiac biostimulator in the heart in the event of dislodgement from a primary fixation site. In some embodiments, the method including the step of entangling an element of the biostimulator within the heart structure at a site within a heart chamber, such entanglement being sufficient to retain the biostimulator within the cardiac chamber. Embodiments of this method may include entangling the biostimulator or an element of the biostimulator within heart structures such as trabeculae in either the left or right ventricle. In another aspect, some embodiments of the invention include preventing escape of the biostimulator into a downstream vascular site, such as the aorta, if preventing escape from the left ventricle, or the pulmonary artery, if preventing escape from the right ventricle.

(25) Some embodiments of the method for retaining a leadless biostimulator in the heart in the event of dislodgement from the primary fixation site include engaging a heart structure-engaging element of the biostimulator within or at a heart structure at as a secondary fixation site within the heart, such engagement being sufficient to retain the biostimulator within the heart chamber. In these embodiments, the engaging step may include positioning a tail connected to a proximal portion of the leadless biostimulator into a second heart chamber. The engaging step may also include restricting the range of movement of the biostimulator in the heart chamber such that the longitudinal axis of the biostimulator cannot be aligned in a direction of downstream blood flow within the heart chamber.

(26) In various embodiment of the method that utilize a proximal tail to engage a heart structure, the first heart chamber is a ventricle and the heart structure includes an atrium. For example, the ventricle can be the right ventricle, in which case the heart structure is the right atrium, and such heart structure may also include the tricuspid valve. As another example, the ventricle can be the left ventricle, in which case the heart structure is the left atrium and/or a semilunar valve. In these embodiments, typically, the engaged heart structure is a heart chamber through which the biostimulator passed prior to being implanted at the primary fixation site in the first heart chamber.

(27) Some embodiments of a method for retaining a leadless biostimulator in a heart chamber in the event of dislodgement from a primary fixation site include expanding an expandable structure to a dimension larger than a diameter of a downstream vascular exit from the heart chamber. In various of these embodiments, the heart chamber is a right ventricle and the downstream vascular structure is a pulmonic valve. Some of these embodiments include coupling an embodiment of an expandable structure to the proximal portion of the leadless biostimulator.

(28) These methods may include, prior to the expanding step, delivering the leadless biostimulator to the heart chamber with the expandable structure in a compressed configuration. After delivery of the LBS to the heart chamber, the method typically includes expanding the expandable structure, and such expanding typically includes bowing struts of the expandable structure from a linear configuration to a bowed configuration.

(29) In some embodiments of the method, expanding the expandable structure comprises expanding the structure in the same heart chamber in which the site of primary fixation is located. The method may further include preventing escape of the biostimulator into a downstream vascular site. Some particular embodiments include preventing escape into the pulmonary artery.

(30) Some embodiments of a method for retaining a leadless intracardiac biostimulator in a heart in the event of dislodgement from a primary fixation site include anchoring the biostimulator with a tether to a secondary anchoring site, the tether being of appropriate length (e.g., sufficiently short) to prevent substantial movement into a downstream vascular from a biostimulator implant site in a heart chamber. In some aspects, anchoring the biostimulator with a tether includes anchoring with a tether of appropriate length to retain the biostimulator within the heart chamber.

(31) In some embodiments, anchoring the biostimulator with a tether includes attaching the tether to an anchor at the secondary fixation site. Such attaching may include attaching the tether to the secondary fixation site with any of a screw, a hook, a clip, a stent, a cage, or a barb.

(32) In various aspects, anchoring the biostimulator to a secondary anchoring site can include anchoring to either an intracardiac site or an extracardial site. In some embodiments, anchoring to an extracardial site includes anchoring to a site on a vessel through which the biostimulator was delivered to the heart. Also, in these embodiments, the anchoring site may be on either an internal or an exterior surface of the vessel.

(33) Some embodiments of a method for retaining a leadless intracardiac biostimulator in a heart in the event of dislodgement from a primary fixation site that include anchoring the biostimulator with a tether to a secondary anchoring site include combining two tethers to form a single tether. Such a method of forming a single combined tether from two original tethers can include inserting a biostimulator attached to a first tether into an entry site in the vasculature, advancing the biostimulator to an intracardial implant site, and implanting the biostimulator at that site, inserting an anchor attached to a second tether into the entry site in the vasculature, advancing the anchor to a secondary anchoring site, and implanting the anchor at that site, and engaging the tether of the biostimulator and the tether of the anchor within a slidable clip at the vascular entry site to form a combined tether. Embodiments of this method may further include adjusting the length of the combined tether by slidably advancing the clip within the vasculature toward secondary anchoring site, and securing the first tether and the second tether at the clip so that no further sliding can occur. More specifically, adjusting the length of the combined tether may include adjusting the length such that there is an appropriate level of slack between the anchoring site and the biostimulator.

(34) In another aspect, rescuing a leadless biostimulator dislodged from its primary fixation site may include a user grasping any portion of a secondary fixation element with a tool, and withdrawing the dislodged biostimulator from the heart chamber in which it was implanted.

(35) In one embodiment, a leadless biostimulator is provided comprising a housing sized and configured to be disposed within a heart chamber of a human, a primary fixation element configured to affix the biostimulator to a heart wall within the heart chamber, and a passive

secondary fixation element configured to prevent an escape of the biostimulator from the heart chamber when the primary fixation element is not affixed to the heart wall, the passive secondary fixation element comprising a tail portion sized and configured to extend from the heart chamber into a second heart chamber without being attached to the human.

(36) In some embodiments, the passive secondary fixation element is sized and configured to be fully disposed within the heart.

(37) In one embodiment, the heart chamber is a right ventricle of the heart and the second heart chamber is a right atrium of the heart, wherein the passive secondary fixation element is sized and configured to extend from the right ventricle through a tricuspid valve into the right atrium.

(38) In some embodiments, the passive secondary fixation element is configured to be grasped by a retrieval tool.

(39) In some embodiments, the passive secondary fixation element is of sufficient stiffness to prevent a dislodged leadless biostimulator from turning or reorienting itself within the heart chamber. The passive secondary fixation element can comprise silicone, polyurethane or any suitable polymer, a flexible metallic core, and/or multifilar windings around the flexible metallic core, for example.

(40) In some embodiments, the passive secondary fixation element comprises an atraumatic tip.

(41) In another embodiment, a leadless biostimulator comprises a housing sized and configured to be disposed within a heart chamber of a human, a primary fixation element configured to affix the biostimulator to a heart wall within the heart chamber, and a passive secondary fixation element configured to prevent an escape of the biostimulator from the heart chamber when the primary fixation element is not affixed to the heart wall, the passive secondary fixation element comprising an expandable structure having a strut, wherein first and second ends of the strut are attached to the housing.

(42) In some embodiments, the passive secondary fixation element is sized and configured to be fully disposed within the heart chamber. The heart chamber, for example, can be the same heart chamber in which the biostimulator is disposed.

(43) In some embodiments, the expandable structure allows deformation into a constrained configuration and a high-fidelity return to a native expanded configuration upon being released from constraint.

(44) In other embodiments, the expandable structure has a minimal cross-sectional linear dimension greater than a diameter of a downstream vascular exit from the heart chamber.

(45) In one embodiment, the strut is disposed parallel to a longitudinal axis of the expandable structure.

(46) In some embodiments, the strut has a native bowed configuration. The bowed configuration can be longitudinally symmetrical or asymmetrical, for example. The asymmetrical bowed configuration can be distributed such that a greater radial diameter is proximally-skewed.

(47) In some embodiments, the strut is configured to be constrainable into a linear configuration without compromising its ability to return to a native configuration.

(48) In some embodiments, the passive secondary fixation device can further comprise a coupling mechanism that couples the expandable structure to the housing. The coupling mechanism can be configured to provide freedom of movement between the expandable structure and the housing. In some embodiments, the coupling mechanism comprises a swivel or a tether.

(49) In one embodiment, the expandable structure comprises a shape memory material, such as Nitinol.

(50) A method of retaining a leadless biostimulator within a human heart of a patient is provided, comprising delivering the leadless biostimulator to a heart chamber, affixing a primary fixation element of the leadless biostimulator to a heart wall of the heart chamber, and extending a passive secondary fixation element of the leadless biostimulator from the heart chamber to a second heart chamber without attaching the secondary fixation element to the patient.

- (51) In some embodiments, the passive secondary fixation element can be sized and configured to be fully disposed within the human heart.
- (52) In one embodiment, the heart chamber is a right ventricle of the heart and the second heart chamber is a right atrium of the heart. The passive secondary fixation element can be extended from the right ventricle to the right atrium through a tricuspid valve.
- (53) In one embodiment, the passive secondary fixation element is of sufficient stiffness to prevent a dislodged leadless biostimulator from turning or reorienting itself within the heart chamber.
- (54) Another method of retaining a leadless biostimulator within a human heart of a patient is provided, comprising delivering the leadless biostimulator to a heart chamber, affixing a primary fixation element of the leadless biostimulator to a heart wall of the heart chamber, and expanding a passive secondary fixation element of the leadless biostimulator within the heart chamber, the passive secondary fixation element comprising a strut wherein first and second ends of the strut are attached to the leadless biostimulator.
- (55) In some embodiments, the passive secondary fixation element is expanded to a dimension larger than a diameter of a downstream vascular structure from the heart chamber.
- (56) In one embodiment, the heart chamber is a right ventricle and the downstream vascular structure is a pulmonic valve.
- (57) In some embodiments, the delivery step further comprises delivering the leadless biostimulator to the heart chamber with the passive secondary fixation element in a compressed configuration.
- (58) In one embodiment, expanding the passive secondary fixation element comprises bowing the strut from a linear configuration to a bowed configuration.
- (59) Embodiments of the invention may further include fixation elements that are redundant, ancillary, or supportive of primary fixation, by, for example, minimizing movement of the biostimulator at the implant site. Such movement may include, for example, undesirable pitch, or yaw, or roll. Some of the embodiments may include rigid elements that are attached or connected to a primary fixation element on one end, and seated into or against heart structure on the other end. Some of these embodiments, which mainly serve in a primary fixation capacity, may further provide a secondary fixation.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B show an embodiment of a leadless biostimulator that has a secondary fixation element in the form of a tail portion that is sized to extend through the tricuspid valve when the biostimulator is placed in the right ventricle. FIG. 1A shows the leadless biostimulator implanted at a primary fixation site in the right ventricle, with the proximal tail extending through the tricuspid valve and into the right atrium.
- (2) FIG. 1B shows the leadless biostimulator embodiment in a situation where it has unfortunately been dislodged from the primary fixation site, but is being retained within the right ventricle because it is prevented from becoming oriented toward the pulmonary valve by its proximal tail.
- (3) FIGS. 2A-2C show an embodiment of a leadless biostimulator with a proximally-connected expandable structure. FIG. 2A shows the leadless biostimulator being delivered to the right ventricle within a delivery device with the struts of the expandable structure in a compressed linear configuration within the confines of the delivery device.
- (4) FIG. 2B shows the leadless biostimulator implanted at a primary fixation site in the right ventricle with the proximally-connected expandable structure in an expanded configuration.
- (5) FIG. 2C shows the leadless biostimulator as implanted with the right ventricle in a contracted state, the ventricle walls compressing the compliant expandable structure.
- (6) FIGS. 3A and 3B show embodiments of a coupling mechanism that connects an expandable

structure to a leadless biostimulator that isolate the LBS from potentially destabilizing force conveyed from the expandable structure to the LBS at its site of primary fixation. FIG. 3A shows a coupling mechanism that includes a swivel.

(7) FIG. 3B shows a coupling mechanism that includes a short flexible tether.

(8) FIG. 3C shows a coupling mechanism that includes a short flexible tether and a swivel mechanism.

DETAILED DESCRIPTION OF THE INVENTION

(9) Leadless biostimulators (LBS's), also known as leadless cardiac pacemakers, for all their advantageous features over conventional pacemakers, could include as part of their profile a risk of loss into the downstream vasculature in the event of dislodgment from their site of primary fixation, were it not for the solution provided by embodiments of this invention. This invention provides various downstream vascular escape prevention methods and assemblies employing, e.g., "secondary fixation" in order to distinguish this form of attachment or fixation from "primary fixation". In this context, primary fixation generally refers to an attachment or fixation of a cardiac pacemaker to an intracardial implant site (or primary fixation site) such that at least one of the electrodes of the biostimulator stably remains in intimate contact with that site on the myocardium. In contrast, secondary fixation generally refers to an element or assembly that retains within the heart chamber a biostimulator that has become loose from its implant site, or prevents the biostimulator from moving any substantial distance into the vasculature downstream from the chamber in which it was implanted, when it has become dislodged.

(10) Retention within the heart chamber thus involves the engagement of one or more secondary fixation elements, at one or more secondary fixation sites. The nature and location of secondary fixation sites may vary in accordance with the nature of the secondary fixation element or the downstream vascular prevention assembly embodiments. Some secondary fixation embodiments include elements that entangle themselves passively within or amongst structural features within the heart chamber, and thus these secondary sites are located within the heart chamber where the device is implanted. These intracardial entangling fixations may be temporary or transient, as the engagement of an entangling element with structure may include sliding or twisting, as examples of transient engagement. In some embodiments or instances, the secondary fixation brought about by an entangling element may effectively become as secure as a typical primary fixation site, either by the effectiveness of entanglement, or by fibrotic process of heart tissue that engages the entangling element. Other embodiments of secondary fixation assemblies, as described herein, may include assemblies comprising an anchor and a tether, the tether connecting the leadless biostimulator to the anchoring site. The anchoring site for these embodiments may be considered the secondary fixation site, and such sites may be intracardial or extracardial. The tether of these embodiments may be composed of any suitable material or mixture of materials, such as, by way of example, single-stranded wire, multi-stranded wire, monofilament suture thread, or multi-stranded suture thread.

(11) Some tether embodiments, as well as other components of secondary fixation elements, may also include an anti-thrombogenic agent to discourage them from becoming a clot-forming nucleus. In some embodiments of the LBS and associated methods of use, the acute phase following implantation is of particular significance in that during that time, the initial period of days or weeks following implantation, the primary fixation becomes more secure, as for example, as a result of the growth of fibrotic tissue envelopes the implant site. Accordingly during that time, the secondary fixation is of particular importance because of the relative vulnerability of the primary fixation. Further, accordingly, in some embodiments it may be appropriate for the tether to include biodegradable materials that degrade over time, after the acute and vulnerable phase has passed. By a similar rationale, it may be appropriate, in some embodiments, for entangling elements or secondary anchors include biodegradable materials.

(12) Secondary fixation embodiments may vary with regard to the extent to which they re-enforce,

assist, support, provide redundancy, or protect the primary fixation method or element. Some embodiments of secondary fixation may serve in one or more of these recited primary fixation-related capacities, either minimally or significantly. Other embodiments for secondary fixation elements or assemblies may provide no substantial contribution to the primary fixation function, and function entirely in their secondary fixation capacity when called upon in the event of failure of the primary fixation.

(13) The U.S. patent publications listed in the background above describe and depict two basic types of primary fixation elements. One embodiment of a primary fixation element is a helix (e.g., FIG. 1A of US 2007/0088418) that may be screwed directly into the myocardium to form a very stable and secure fixation. The screwable helix approach to primary fixation may be considered “active” in that it entails a screwing action to seat it, and it is at least to some extent invasive of the myocardium. A second embodiment of a primary fixation element described therein includes a small set of tines (e.g., FIG. 1B of US 2007/0088418) that may be used alone or in combination with a screwable helix, and which are designed particularly to establish lateral stability on the myocardial surface. The primary fixating tines may be considered relatively “passive”, in comparison to the actively engaging screwable helix, as the engagement of the tines to the surface does not involve a screwing action, and the engagement is minimally invasive of the surface of the myocardium. Primary fixating tines typically do not extend or do not substantially extend beyond the diameter profile of the biostimulator, typically being less than 5 mm in length. Further, depending on the embodiment and the nature of the engagement of the primary fixating site, the tines may be directed at an angle that varies between proximal and distal. The fixation provided by these tines may serve as a stand-alone fixation element, but may also be used in conjunction with a helix, in which case they may be understood to be a redundant, back-up, or supportive form of primary fixation. Both types of primary fixation elements are subject to fibrotic overgrowth, as mentioned in the background, which further supports the fixation of the LBS at the attachment site.

(14) The secondary fixation elements described herein perform a fail-safe function by, after failure of primary fixation, preventing loss of a dislodged LBS from a ventricle in which it is implanted, and they may further, in some embodiments, support stability of the LBS at the implant site. For example, if an LBS implanted in the right ventricle were to dislodge and exit the ventricle, it would leave through the pulmonic valve and lodge in the lungs. If an LBS implanted in the left ventricle were to exit the ventricle, it would enter the aorta and move into the general circulation, or the brain. A function of secondary fixation is to prevent occurrence of these catastrophic events should primary fixation fail. Some embodiments of the secondary fixation elements effectively retain a dislodged LBS within the ventricle, and other embodiments may allow exit from the ventricle for a very short distance but stop any substantial downstream movement. Dislodgment or detachment of an LBS from its implant site, even with loss from the ventricle and adverse downstream consequences being prevented, is nevertheless a serious medical emergency, and the loosed LBS needs to be retrieved. Thus, another benefit and function of the secondary fixation element is that it may contribute to the feasibility of a retrieval procedure, by providing an element easily graspable by a retrieval tool.

(15) As with primary fixation elements, secondary fixation elements may be active (or actively-applied) or passive (or passively-engaging). Active secondary fixation elements include a tether that connects the LBS to an anchor at a secondary site, the anchor being a secure attachment made by active engagement of a portion of the heart or engagement at an extracardial site. Passive secondary fixation embodiments include elements that hook, snag, or otherwise entangle within intrachamber structural features of the heart, but they are substantially non-invasive of heart structure, nor are they actively seated during implantation of the LBS. Anatomical heart structure in the chamber in which the elements entangle includes connective tissue structures generally referred to as trabeculae carneae that are prominent in ventricles, and may also include ridges in the myocardium, and may also include tissue with a mix of fibrous and muscular tissue. Trabeculae

carneae may be referred to simply as trabeculae in the cardiac context; the structures are attached to the chamber wall and vary in form, appearing as ridges, flaps, and cords.

(16) Embodiments of passive secondary fixation elements or entangling elements are typically closely associated with the body of the LBS, i.e., they are integral with the body of the LBS, directly attached to it, or mounted on a rotatable collar encircling the LBS. A typical embodiment of an entangling element is a set of one or more tines projecting outwardly from the body of the LBS, as described and depicted in detail below. In some embodiments, tines may include features that further provide engaging or particularly entangleable features, such as hooks, typically atraumatic hooks, or linked elements, such as for example, serial structures threaded together, or linked as in a chain. Tines may assume various forms; they may be straight or curved, they may project at various angles from the leadless biostimulator, and they may have a collapsible bias. Such collapsibility is advantageous for several reasons. In one aspect collapsibility reflects a flexible and compliant quality of the tines which is compatible with them being a structure that does not interfere with primary fixation. Further, the collapsibility has a bias that is typically proximally directed; this bias is consistent with the configuration of the landscape of the heart chamber that surrounds the primary attachment site. Collapsibility also provides for a structure that folds easily and closely around the body of the leadless biostimulator, which is a property advantageous for being accommodated by a delivery device, and further is compatible with being enclosed within a soluble capsule for deployment, and expanding outward to post-deployment configuration after dissolution of the soluble capsule. Typically, embodiments of tines project outwardly beyond the diameter of the leadless biostimulator to which they are attached, and typically, such tines are about 5 mm in length or longer.

(17) Entangling elements may be attached to the LBS housing at any point along the body from proximal end to distal end, although they are generally not located at the distal-most point, because that locale is typically the location of a primary fixation element. The rotatable collar may be understood as a mount upon which tines may rotate around the main axis of the LBS body, or, from the complementary perspective, as a collar within which the LBS body may rotate. Rotation of the LBS body within the collar allows the body to turn as a screw, a movement that embeds a primary fixating helix into the myocardium while allowing the tines to come to rest as they encounter obstructing trabeculae in which they entangle.

(18) In another embodiment of secondary fixation, the secondary fixation element, a downstream vascular escape prevention assembly that is adapted to prevent an escape of the biostimulator in the event of it being dislodged from the primary fixation site, can be a passive heart structure-engaging element. Such structure may be within the same chamber as the primary fixation site, or it may be located elsewhere within the heart. As an example, a passive heart engaging element may engage or be engaged by a valve or a chamber that is upstream from the site of primary fixation. For example, an LBS implanted in the right ventricle may be engaged within the tricuspid valve and the right atrium.

(19) FIGS. 1A and 1B show an embodiment of a leadless biostimulator that has a secondary fixation element in the form of a tail portion that is sized to extend through the tricuspid valve when the biostimulator is placed in the right ventricle. FIG. 1A shows an embodiment of a leadless biostimulator **10** implanted in the right ventricle **102** of a heart **100**, the LBS having a tail **44** extending from its proximal end that extends proximally through the tricuspid valve **108** and into the right atrium **103**. As shown in FIGS. 1A-1B, the tail **44** is not affixed or attached to patient. As also shown in FIG. 1B, in the rare and unfortunate event of dislodgement of the LBS from its primary fixation site, the wire tail **44** is of sufficient stiffness that it prevents the loosed LBS from turning or reorienting itself within the heart chamber to the extent that it can turn and be aligned to enter the semilunar valve **109**. By not being able to pass through the semilunar valve **109** and enter the pulmonary artery **138**, the LBS necessarily remains within the right ventricle. The distal end of the tail remains projecting distally into the right atrium, where it can be grasped by a retrieval tool

and retrieved. The tail portion **44** may include features that discourage attachment and colonization of its surface by cells of either blood or cardiac origin. Exemplary embodiments of the tail may include silicone or polyurethane or any suitable polymer, and may include a flexible metallic core. Stiffness can be added to a metallic core by the inclusion of multifilar windings around the core. In some embodiments, the tail **44** has an atraumatic tip adapted such that its contact with heart structure is benign, causing no irritation or injury. The tail **44** may additionally be adapted or configured to facilitate easy capture by a retrieval device.

(20) FIGS. 2A-2C show an embodiment of a leadless biostimulator **10** with a proximally-connected expandable structure. In some embodiments, the expandable structure includes Nitinol, which provides the structure a superelastic quality that allows deformation into a constrained configuration and a high-fidelity return to a native configuration upon being released from constraint. FIG. 2A shows the leadless biostimulator being delivered to the right ventricle within a delivery device **200** with the struts of the expandable structure in a compressed linear configuration within the confines of the delivery device. After the device is successfully implanted at a primary fixation site in the right ventricle, the delivery device is withdrawn as it ejects the LBS **10**. As shown in FIG. 2B as the expandable structure **52** is released from the constraint provided by the delivery device, the struts **53** return to their native bowed configuration, giving the expandable device as a whole its expanded configuration. As shown in FIGS. 2B-2C, the struts can comprise a looped shape, wherein first and second ends of the struts are attached to the housing of the biostimulator. Although the expandable structure is shown with a plurality of struts in FIGS. 2B-2C, in some embodiments, the expandable structure can comprise a single strut.

(21) Depending on the orientation of the expandable structure **52** within the heart chamber, and the relative dimensions of the structure and the host chamber, the structure may contact the heart chamber wall to varying degree, or it may be disposed in such a way that there is minimal contact with the heart chamber wall. In some cases, the walls of the chamber may compress the expandable structure as the ventricle contracts during a heart beat. This event is depicted in FIG. 2C, which shows the leadless biostimulator as implanted with the right ventricle in a contracted state, the ventricle walls compressing the compliant expandable structure. As the ventricle walls contract, the struts **53**, which are highly compliant, compress and do not interfere with the movement of the heart muscle.

(22) FIGS. 3A-3C show details of embodiments of a coupling mechanism **55** that couples the leadless biostimulator **10** and the expandable structure **52** that allows a freedom of movement of the expandable structure with respect to the biostimulator. This freedom of movement provides several advantages to the long term stability of the LBS at its primary implantation site. In one aspect, the freedom of movement isolates or insulates the LBS from destabilizing force that could otherwise be conveyed to it by movement of the coupled expandable structure. In another aspect, the freedom of movement allows the expandable structure to maintain a position within the heart chamber that least interferes with the wall of the heart chamber, and more particularly, least interferes with contraction of the chamber as the heart beats. In yet another aspect, the expandable structure provides a net stabilizing effect on the primary fixation by limiting the movement of the LBS in response to the movement of blood and the movement of the cardiac muscle at the site of fixation to less than it would be absent such a coupled structure.

(23) FIG. 3A shows a coupling mechanism **55** that includes a swivel that allows the expandable structure to rotate within the ventricle to a position where it encounters the least interaction with portions of the ventricle wall. By virtue of the swivel, the LBS is insulated from torque that could otherwise be applied to the LBS at its site of implantation, and which could destabilize the primary fixation. FIG. 3B shows a coupling mechanism **55** that includes a short flexible tether that allows the expandable structure to wobble within the ventricle with respect to the LBS to which it is coupled. Again, the coupling mechanism thus insulates the LBS from destabilizing force from the movement of the expandable structure which could threaten the integrity of the primary fixation.

Other embodiments of a coupling mechanism 55 (not shown) may include both tethering and swiveling features. FIG. 3C shows a coupling mechanism that includes a short flexible tether and a swivel mechanism, thus incorporating the features of a swivel and short tether, as just described. (24) As for additional details pertinent to the present invention, materials and manufacturing techniques may be employed as within the level of those with skill in the relevant art. The same may hold true with respect to method-based aspects of the invention in terms of additional acts commonly or logically employed. Also, it is contemplated that any optional feature of the inventive variations described may be set forth and claimed independently, or in combination with any one or more of the features described herein. Likewise, reference to a singular item, includes the possibility that there are plural of the same items present. More specifically, as used herein and in the appended claims, the singular forms “a,” “and,” “said,” and “the” include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation. Unless defined otherwise herein, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. The breadth of the present invention is not to be limited by the subject specification, but rather only by the plain meaning of the claim terms employed.

Claims

1. A leadless biostimulator comprising: a housing sized and configured to be disposed within a heart chamber of .[a.]. .Iadd.an adult .Iaddend.human; a primary fixation element configured to actively affix the biostimulator to a heart wall within the heart chamber; and an expandable structure connected to a proximal portion of the housing, the expandable structure being adapted to assume a compressed linear configuration during delivery and an expanded configuration after delivery, wherein the expanded configuration of the expandable structure has a cross-sectional dimension that is greater than a diameter of a vascular exit pathway from the heart chamber so that expansion of .Iadd.the .Iaddend.expandable structure confines the leadless biostimulator and the expandable structure to the heart chamber.Iadd., wherein the expandable structure further comprises a coupling mechanism that couples the expandable structure to the housing to allow for movement of the expandable structure with respect to the housing.Iaddend..
2. The leadless biostimulator of claim 1 wherein .Iadd.the .Iaddend.expandable structure comprises a plurality of struts.
3. The leadless biostimulator of claim 2 wherein the plurality of struts comprise a looped shape.
4. The leadless biostimulator of claim 2, wherein the plurality of struts has an asymmetrical bowed configuration such that a greater radial diameter is proximally-skewed.
5. The leadless biostimulator of claim 2, wherein the plurality of struts is configured to compress when the heart chamber contracts during a heart-beat so that the plurality of struts do not interfere with movement of heart muscle of the heart chamber.
6. The leadless biostimulator of claim 1 wherein the heart chamber is a right ventricle of the heart and the vascular exit pathway from the heart chamber is a pulmonic .[.value.]. .Iadd.valve.Iaddend..
7. The leadless biostimulator of claim 1 wherein the expandable structure comprises nitinol.
8. The leadless biostimulator of claim 1, wherein the expandable structure further comprising a coupling mechanism that couples the housing to the expandable structure to allow for movement of the expandable structure with respect to the housing..].
9. The leadless biostimulator of claim .[.8.]. .Iadd.1.Iaddend.wherein the coupling mechanism comprises a swivel, wherein the swivel is configured to insulate the biostimulator from torque when the biostimulator is affixed to the heart wall with the heart chamber.

10. The leadless biostimulator of claim 1, further comprising a coupling mechanism that couples the housing to the expandable structure to allow for movement of the expandable structure with respect to the housing, wherein the coupling mechanism isolates the housing from destabilizing force that could otherwise be conveyed to the housing by movement of the expandable structure.

11. The leadless biostimulator of claim 1, further comprising a one or more soluble coverings configured to encapsulate at least one of the primary fixation element and the secondary fixation element.

12. The leadless biostimulator of claim 1, further comprising a one or more soluble coverings configured to encapsulate at least one of the primary fixation element and the secondary fixation element.

13. The leadless biostimulator of claim 1, further comprising a one or more soluble coverings configured to encapsulate at least one of the primary fixation element and the secondary fixation element so that the secondary fixation element is secured in the collapsed linear configuration.

14. The leadless biostimulator of claim 1, wherein the expandable structure has a stiffness configured to prevent the biostimulator from turning or reorienting within the heart chamber when the biostimulator is dislodged from the heart wall.

15. The leadless biostimulator of claim 1, wherein the heart chamber is a right ventricle of the heart and the vascular exit pathway from the heart chamber is a pulmonic artery.

16. The leadless biostimulator of claim 1, wherein the heart chamber is a right ventricle of the heart and the vascular exit pathway from the heart chamber is a semilunar valve, and wherein the expanded configuration of the expandable structure is configured not to pass through the semilunar valve so that the biostimulator does not enter the pulmonary artery.

17. A method of retaining a leadless biostimulator within a human heart of a patient, comprising: confining an expandable member of the leadless biostimulator within a delivery device; delivering the leadless biostimulator to a heart chamber with the delivery device; actively affixing a primary fixation element of the leadless biostimulator to a heart wall of the heart chamber; and expanding the expandable member within the heart chamber to have a cross-sectional linear dimension greater than a downstream vascular exit from the heart chamber such that the expandable member is confined to the heart chamber and the leadless biostimulator is confined to the heart chamber by the expandable member, wherein the expandable member further comprises a coupling mechanism that couples the expandable member to a housing of the biostimulator to allow for movement of the expandable member with respect to the housing.

18. The method of claim 17 wherein the heart chamber is a right ventricle of the heart and wherein the downstream vascular exit from the heart chamber is pulmonary artery.

19. The method of claim 17, wherein expanding the expandable member within the heart comprises bowing struts of the expandable member from a linear configuration to a bowed configuration.

20. A leadless biostimulator comprising: a housing sized and configured to be disposed within a heart chamber of an adult human; and a plurality of fixation features including a primary fixation element configured to actively affix the biostimulator to a heart wall within the heart chamber, and a secondary fixation element including a plurality of struts that collectively form an expandable structure connected to a proximal portion of the housing, the plurality of struts of the expandable structure being adapted to assume a compressed linear configuration during delivery and an expanded bowed configuration after delivery, wherein the expanded bowed configuration of the expandable structure has a proximally-skewed cross-sectional dimension that is greater than a diameter of a vascular exit pathway from the heart chamber so that expansion of the expandable structure confines the leadless biostimulator and the expandable structure to the heart chamber, wherein the expandable structure further comprises a coupling mechanism that couples the expandable structure to the housing to allow for movement of the expandable structure with respect to the housing.

21. The leadless biostimulator of claim 20, wherein the plurality of struts are configured to

compress when the heart chamber contracts. .Iaddend.

.Iadd.22. A leadless biostimulator comprising: a housing sized and configured to be disposed within a heart chamber of an adult human; and a plurality of fixation features including a primary fixation element configured to actively affix the biostimulator to a heart wall within the heart chamber, and a secondary fixation element having an expandable structure connected to a proximal portion of the housing, the expandable structure including a plurality of tines mounted on a collar encircling the housing and being adapted to assume a compressed linear configuration during delivery and an expanded configuration after delivery, wherein the expanded configuration of the expandable structure is a native outwardly bowed configuration having a cross-sectional dimension that is greater than a diameter of a vascular exit pathway from the heart chamber so that expansion of the expandable structure confines the leadless biostimulator and the expandable structure to the heart chamber. .Iaddend.

.Iadd.23. The leadless biostimulator of claim 22, wherein the heart chamber is a right ventricle of a heart and the vascular exit pathway from the heart chamber is a pulmonary valve. .Iaddend.

.Iadd.24. A leadless biostimulator comprising: a housing having a longitudinal axis and being sized and configured to be disposed within a heart chamber of an adult human; and a plurality of fixation features including a primary fixation element configured to actively affix the biostimulator to a heart wall within the heart chamber, and a secondary fixation element having an expandable structure connected to a proximal portion of the housing, the expandable structure including a plurality of struts disposed parallel to the longitudinal axis and being adapted to assume a compressed linear configuration during delivery and an expanded configuration after delivery, wherein the expanded configuration of the expandable structure has a cross-sectional dimension that is greater than a diameter of a vascular exit pathway from the heart chamber so that expansion of the expandable structure confines the leadless biostimulator and the expandable structure to the heart chamber, wherein the expandable structure further comprises a coupling mechanism that couples the expandable structure to the housing to allow for movement of the expandable structure with respect to the housing. .Iaddend.

.Iadd.25. The leadless biostimulator of claim 24, wherein the expandable structure comprises nitinol. .Iaddend.
